LabOS | Scientific Skills, Connected Instruments

Agents decide • skills guide workflows • MCP connects tools • local workers execute

Operator workspace: objective, setup, review and authorization

Cloud LLM API • remote

ONE CENTRAL SERVER / LABOS RUNTIME

Scientific reasoning, reusable methods, tool access, model routing and persistent execution state

Knowledge & memory

SEMANTIC KNOWLEDGE

Literature / manuals / local code
Setup definitions + calibrations

EPISODIC EXPERIMENT MEMORY

Runs / artifacts / faults / decisions
Versioned evidence + provenance

Retrieve / persist

Scientific agent runtime

LabOS control agent / task supervisor

Planning agent | **Analysis agent** | **Diagnostics agent**

Observe → frame → select skills → delegate → review → update

LLM

Model runtime / routing

Cloud API • preferred
Remote inference backend

Local LLM • on this server
Fallback inference backend

Internet / API / quota failure → local model
Context + tool permissions stay consistent

Load skill

Skill library

Brainstorming | **Experiment design**
Calibration + QC | **Instrument integration**
Analysis + reporting | **Troubleshooting**

SKILL.md + references / scripts
Discover → load on demand → follow workflow

Structured results → evidence

Discover tools / invoke capability

Tool access / MCP host + clients

Knowledge MCP
Search / read / retrieve

Analysis MCP
Fit / compare / visualize

LabOS Control MCP
Request / status / cancel

Connector services expose tools, resources and prompts through a common interface.
Additional connectors: papers, files, code, reporting and software integration

Capability + setup registry

PC identity → device bindings
Operations + parameter schemas
Units / limits + calibration
Shared resources + reservations
Run / stage / point context

PC roles follow registered capabilities

Results / status / artifact handles

LabOS control tools → validated execution

Execution authority / deterministic stage coordinator

Validate units + limits | **Reserve resources** | **Join READY states** | **Track command outcome** | **Qualify + archive data**

PC-INITIATED HTTP(S) • commands, heartbeat, results and artifacts

LAN / VPN / central HTTPS • no SSH dependency

Command payloads ↓

Readbacks + evidence ↑

PC i / instrument host

Local execution, vendor environment and data

PC Agent → bounded drivers → device sessions

Watchdog • command journal • raw-data outbox

SDK / VISA / USB / serial / LAN

Local instruments / controllers / detectors

Device 1 • Device 2 • … • Device m

PC j / instrument host

Local execution, vendor environment and data

PC Agent → bounded drivers → device sessions

Watchdog • command journal • raw-data outbox

SDK / VISA / USB / serial / LAN

Local instruments / controllers / detectors

Device 1 • Device 2 • … • Device m

PC N / instrument host

Local execution, vendor environment and data

PC Agent → bounded drivers → device sessions

Watchdog • command journal • raw-data outbox

SDK / VISA / USB / serial / LAN

Local instruments / controllers / detectors

Device 1 • Device 2 • … • Device m

Representative PCs: i, j, …, N. Any PC may host any registered instrument set.

AI inference is central; PC-side LLMs are optional.

Skills = reusable methods | MCP = tool interfaces | Coordinator = execution authority | PC Agent = local device I/O